CS 594: Vector Databases – Programming Project

Fall 2026, A. Asudeh

Final Submission Deadline: Monday, December 7, 2026, at 11:59 PM

Project Overview

The programming project focuses on developing a Vector Database and Retrieval-Augmented Generation (RAG) system for pathology images.

Pathology images contain rich visual information but often lack detailed descriptive metadata. The goal is to investigate whether existing image descriptions, combined with vector similarity search and RAG, can produce useful descriptive information about pathology images.

The recommended dataset is PathVQA (feel free to use any additional real-world/synthetic datasets):

https://huggingface.co/datasets/flaviagiammarino/path-vqa

PLIP is a suggested embedding model:

https://github.com/pathologyfoundation/plip

Teams are responsible for selecting their implementation approach and evaluating the resulting system.

GitHub Repository

All programming projects will be maintained in our course GitHub organization:

https://github.com/UIC-CS594-Fall2026

Each team must create one **public repository** using the provided template:

https://github.com/UIC-CS594-Fall2026/project-template

1. Accept your invitation to the course GitHub organization.
2. Open the template repository and select **Use this template** → **Create a new repository**.
3. Select `UIC-CS594-Fall2026` as the owner.
4. Choose a meaningful repository name and set its visibility to **Public**.
5. Add your teammates as collaborators.

All project code must be committed to this repository. Teams should commit their work regularly so that the repository reflects the development of the project.

Required Documentation

Each team must provide detailed documentation in its GitHub repository. The documentation should allow someone unfamiliar with the project to understand, install, run, and evaluate the system.

At a minimum, include:

1. **Team information:** Names of team members and their individual contributions.
2. **Project description:** The problem being addressed, the objectives of the system, and its main functionality.
3. **Datasets and system functionalities:** Describe the datasets you used, clarify any assumptions made, and discuss the functionalties the system provides.
4. **System architecture:** A clear explanation of the components and their interactions, including embedding generation, vector storage and retrieval, and RAG.
5. **Implementation details:** The vector database, embedding models, algorithms, libraries, and other tools used. Explain important design decisions and implementation choices.
6. **Installation and execution:** Dependencies, setup instructions, configuration, and commands needed to reproduce the demonstration.
7. **Evaluation:** Describe the dataset, experimental setup, evaluation methodology, and results. Discuss what worked, what did not, and the limitations of the system.
8. **AI usage disclosure:** A detailed account of whether and how AI tools were used, as specified below.

Documentation may be organized in the main `README.md` or in additional Markdown files under a `docs/` directory.

AI Usage Disclosure

Every team must explicitly document its use of AI tools, including LLMs, during the project.

Create a separate file named `AI_USAGE.md` in your repository.

For each AI tool used, document:

- The tool and model used, if known.
- The specific tasks for which it was used.
- Which parts of the code or documentation, if any, were affected.
- How the generated suggestions were verified or modified.

Be specific. Statements such as "We used ChatGPT for coding" or "AI was used for debugging" are not sufficient. Explain what assistance was requested and where it was applied.

If no AI tools were used, explicitly state this in `AI_USAGE.md`.

The course AI usage policy remains in effect. AI assistance is permitted for understanding concepts and syntax-level questions. LLM-generated algorithm design and vibe coding are not permitted.

Failure to disclose AI usage is an academic integrity violation. Disclosure does not make otherwise prohibited AI usage permissible.

Recorded Video Demonstration

Each team must submit a recorded video demonstrating its completed system.

The video should:

1. Briefly introduce the project and team.
2. Explain the system architecture and main implementation choices.
3. Demonstrate the working system using representative inputs and outputs.
4. Present the main experimental results.
5. Discuss the system's limitations.

The demonstration must show the actual implementation running. Slides alone are not sufficient.

The video should be accessible through a link included in the repository's `README.md`. Verify that the instructor can access the video without requesting additional permissions.

Final Submission

Deadline: Monday, December 7, 2026, at 11:59 PM (Chicago time).

Each team must submit its GitHub repository URL through Canvas by the deadline.

The repository must contain:

1. Complete project source code.
2. Detailed project documentation.
3. The `AI_USAGE.md` disclosure file.
4. A link to the recorded video demonstration.

Only one submission is required per team.

The submitted repository must be public and accessible at the deadline. All required materials must be available by that time.

Academic Integrity and Late Submissions

All submitted work must be original and must comply with the course academic integrity and AI usage policies.

The course late-submission penalty of 20% per day applies. "'